Pearson
BTEC

Examiners' Report
Lead Examiner Feedback

January 2022

Pearson BTEC Nationals
In Applied Science (31617H)
Unit 1: Priciples and Applications of Science I

Edexcel and BTEC Qualifications

Edexcel and BTEC qualifications come from Pearson, the world's leading learning company. We provide a wide range of qualifications including academic, vocational, occupational and specific programmes for employers. For further information visit our qualifications website at http://qualifications.pearson.com/en/home.html for our BTEC qualifications.

Alternatively, you can get in touch with us using the details on our contact us page at http://qualifications.pearson.com/en/contact-us.html

If you have any subject specific questions about this specification that require the help of a subject specialist, you can speak directly to the subject team at Pearson. Their contact details can be found on this link:
http://qualifications.pearson.com/en/support/support-for-you/teachers.html

You can also use our online Ask the Expert service at https://www.edexcelonline.com You will need an Edexcel Online username and password to access this service.

Pearson: helping people progress, everywhere

Our aim is to help everyone progress in their lives through education. We believe in every kind of learning, for all kinds of people, wherever they are in the world. We've been involved in education for over 150 years, and by working across 70 countries, in 100 languages, we have built an international reputation for our commitment to high standards and raising achievement through innovation in education. Find out more about how we can help you and your learners at: www.pearson.com/uk

January 2022
Publications Code 31617H_2201_ER

Grade Boundaries

What is a grade boundary?

A grade boundary is where we set the level of achievement required to obtain a certain grade for the externally assessed unit. We set grade boundaries for each grade, at Distinction, Merit and Pass.

Setting grade boundaries

When we set grade boundaries, we look at the performance of every learner who took the external assessment. When we can see the full picture of performance, our experts are then able to decide where best to place the grade boundaries – this means that they decide what the lowest possible mark is for a particular grade.

When our experts set the grade boundaries, they make sure that learners receive grades which reflect their ability. Awarding grade boundaries is conducted to ensure learners achieve the grade they deserve to achieve, irrespective of variation in the external assessment.

Variations in external assessments

Each external assessment we set asks different questions and may assess different parts of the unit content outlined in the specification. It would be unfair to learners if we set the same grade boundaries for each assessment, because then it would not take accessibility into account.

Grade boundaries for this, and all other papers, are on the website via this link:
http://qualifications.pearson.com/en/support/support-topics/results-certification/grade-boundaries.html

Awarding BTEC qualifications in 2022

Ofqual has set out their plans for awarding qualifications in 2022 and intend to return to a normal, pre-pandemic, approach to grading standards over by 2023. They have confirmed that 2022 will be a transition year, to reflect that we are in a pandemic recovery period and students' education has been disrupted.

Our guiding principle and approach to awarding BTEC qualification results in 2022 will be to ensure parity in relation to the approach being taken for GCSE and A level learners. BTEC courses have a different structure and design to academic qualifications - BTECs are modular qualifications (with assessments taking place throughout the course) compared to GCSEs and A levels which are linear (assessed and awarded at the same time at the end of the year), and therefore our approach needs to be different.

In 2022 we will return to the usual method of calculating BTEC qualification results, however adaptations including, U-TAGs and reduced internal assessment, are in place to provide a comprehensive package of support for students.

The basis of our awarding approach to BTECs this year is to ensure it is as fair as possible for all learners. We will use a range of evidence to set grade boundaries for the external units. Part of this evidence will be to closely monitor learner performance in all assessments that contribute to learners' final qualification grade, to ensure parity with A level and GCSEs.

Further information can be found on our website and via our Social Media channels.

Unit 1: Principles and Applications of Science I

Grade	Unclassified	Level 3			
		N	P	M	D
Boundary Mark	0	11	22	40	58

Introduction

Biology

Structure and functions of cells and tissues.

Medical professionals need to understand the structure and workings of cells. They build on this knowledge to understand how the body stays healthy as well as the symptoms and causes of some diseases. This allows them to diagnose and treat illnesses. The study of bacterial prokaryotic cells gives an understanding of how some other diseases are caused and can be treated.

Introduction to the Overall Performance of the Unit

Structure and functions of cells and tissues.

Time management and the depth of knowledge expected on some key areas of the specification seem to be the biggest issue for learners. The additional guidance document exemplifies the expectations of the specification. Centres need to fully prepare learners for the exam by practising exam technique, especially in relation to reading the question carefully and not repeating the stem of the question. Learners should also be taught that when they have answered the question to reread their response in order to ensure that the question set has been addressed in the answer they have given, and that they have used appropriate scientific knowledge and vocabulary. Exam technique and the understanding of the command verbs is continuing to improve. It is beneficial for learners to appreciate the requirement of each of the command verbs so that they can target their response and provide appropriate credit worthy answers. Explain is the command verb which seems least understood, especially for a 2 mark question. Learners tend to make an identification point without a linked expansion for full marks, or learners give a number of points, suggesting an imbalance of time spent on short questions. Learners should show that they understand the relationship between the structure and function of cells and tissues. Learners still struggle to recall definitions and find it difficult to provide links between different related aspects of the specification.

Introduction

Chemistry

This was the eight sitting of the Level 3 Applied Science Unit 1 Chemistry section and the fifth time the paper has been sat separately to the other disciplines that makes up the unit.

Introduction to the Overall Performance of the Unit

This sitting of the Level 3 Applied Science Unit 1 Chemistry section was well received by learners, with many making good attempts at all questions and fewer leaving blank spaces than in previous series.

Successful learners were able apply their Chemistry to new situations and contexts. They were able to perform the given calculations using clear and logical working and could understand and use scientific vocabulary in the correct context. Their knowledge of basic scientific terminology allowed them to access questions and apply ideas to new situations.

Introduction

Physics

The examination was set to allow learners to show their knowledge and understanding of waves in the context of communication and to use their knowledge to interpret and analyse the use of waves in different applications.

Introduction to the Overall Performance of the Unit

Mathematical skills have improved, with the majority of learners showing substitution into an equation and gaining at least one mark for this even if the rest of the calculation was not completed. However, rearranging equations, squaring and square roots are still found to be challenging with some learners not understanding the nomenclature used in equations. Most learners were able to determine amplitude from a graph but found it more difficult to decide on which length constituted a wavelength. Learners also need to find some way of remembering the difference in particle motion for transverse and longitudinal waves as this was often confused.

There was limited knowledge of the process used by a computer to access the internet many learners were unaware of the existence of the router.

The relationship between frequency and wavelength for sound in the same length closed and open pipes was not appreciated by many students although the diagrams for the pipes showed the part of the wave in each pipe. The majority of learners were able to access the extended response question with a few giving answers that demonstrated a good knowledge of relevant information which was well reasoned and logically presented.

Individual Questions

Biology

Structure and functions of cells and tissues.

Q1(a)

This question was answered well by the majority of learners. Learners should be able to recognise cell components from photomicrographs and electron micrographs.

(a) Identify the cell component labelled **X** in Figure 1.

☐ **A** Golgi apparatus

☒ **B** Nucleus

☐ **C** Vacuole

☐ **D** Vesicle

Q1(b)

Most learners were able to provide the correct answer. The most common errors seen were multiplying instead of dividing, or a power of ten error.

Lead Examiner comment: *This response was awarded 2 marks. The learner has shown their working out, which is good practice.*

(2)

$$M = \frac{I}{A}$$

$$\cancel{500 = \frac{x}{4100}} \qquad 500 = \frac{4100}{x}$$

$$\cancel{x =} \qquad 500x = 4100$$
$$x = 8.2$$

actual width = 8.2 µm

Lead Examiner comment: *This response was awarded 1 mark. The answer is incorrect so if the learner has provided their working out, the work can be investigated for any creditworthy content. A correct substitution is seen and so MP1 is given. The evaluation is incorrect and so no further marks are given.*

(2)

$$m = \frac{\text{size of image}}{\text{size of real object}}$$

$$500 = \frac{41000}{?} = 20\,50000 \div 1000 = 2050\,\mu m$$

actual width = 2050 µm

Q1(c)(i)

Most learners were able to achieve the marking point for this question. Commonly seen incorrect answers included ‘lungs’, ‘nose’ or ‘throat’, and oesophagus. A very small number of learners referred to the fallopian tube, which would have been a correct response.

Lead Examiner comment: *This response was awarded 1 mark. The learner has given trachea as their response and so is awarded 1 mark.*

(1)

Trachea

Lead Examiner comment: *This response was awarded 0 marks. Lungs is a common response, but unfortunately too vague at Level 3 to be credited as the lungs contain other tissues which are not ciliated epithelial cells.*

(1)

Lungs

Q1(c)(ii)

The majority of learners were able to achieve at least one of the marks available for this question, but many had misconceptions as to the purpose of cilia. A number of learners suggested that the ciliated cells produced mucus, suggesting a confusion with goblet cells. A significant number of answers confused cilia with villi, referring to increasing the surface area for absorption. Others conflated cilia with mucus, writing about trapping bacteria or particles without reference to movement. Some learners referred to cilia being involved in transport across the cell membrane, aiding cell signalling or otherwise directly help cells carry out their function. Learners should be able to show that they understand the relationship between the structure and function of cell components.

Lead Examiner comment: *This response was awarded 2 marks. The learner has stated that the cilia "move things" which is just an acceptable description for MP1.*
They then go on to name bacteria and particles as the substances moved which is correct for MP2.

(2)

The cilia function is move thing offs prevent the bacterial and particle in to the lung by trachea or

Lead Examiner comment: *This response was awarded 1 mark. Learner has not given the idea of movement for MP1 - we are looking for words such as moves, brushes away, removes etc to give the correct context of transport The learner is given MP2 as they have correctly named bacteria as the substance. The learner would not be given a mark for their use of "mucus" alone in this response as it is given in the wrong context - that of production. Responses along the lines of "traps bacteria" were very common -and these responses were only given MP2.*

It protects our inner linning and lungs against the bacteria. All the bacteria gets collected and mucus forms and only clean air goes in.

(Total for Question 1 = 6 marks)

Q2(a)

The majority of leaners were able to provide the correct answer. Learners should be able to recognise cell components from photomicrographs and electronmicrographs.

(a) Identify the cell component labelled **Y** in Figure 2.

☐ **A** amyloplast

☐ **B** chloroplast

☐ **C** mitochondrion

☒ **D** ribosome

Q2(b)

This question was not answered. From the incorrect responses, no one answer appears to have been favoured, implying many learners employed guesswork. However, if the learner cannot make an educated response or guess than the learner should be encouraged to select a response to multiple choice questions, rather than leaving it blank, as the paper is not negatively marked for incorrect responses and so the learner only stands to gain from an attempt.

(b) After amino acids have been assembled into a protein, they enter the RER.

Some of these proteins leave the cell.

Identify the route the proteins take from the RER to the plasma membrane.

(1)

☒ **A** RER → Golgi apparatus → mitochondria → transport vesicles → plasma membrane

☒ **B** RER → lysosomes → transport vesicles → Golgi apparatus → plasma membrane

☒ **C** RER → transport vesicles → Golgi apparatus → mitochondria → plasma membrane

☒ **D** RER → transport vesicles → Golgi apparatus → transport vesicles → plasma membrane

Q2(c)

A minority of learners were able to identify the synthesis of lipids or the storage of lipids and were awarded 2 marks for this question.
Many learners seemed to confuse the role of the SER and the RER and referenced proteins in their response. Some were confused with the role of the Golgi body. A common misconception was that smooth endoplasmic reticulum is involved in exocytosis

Some detail to ensure that the learner was referring to the SER was required for the marks. Producing and storing lipids, cholesterol and phospholipids or carbohydrate metabolism were all credit worth responses.
A few learners stated a structural feature of SER e.g. no ribosomes present rather than a function. Learners should read the question carefully to avoid this error.

Lead Examiner comment: *This response was awarded 2 marks. Correct responses can be grouped into one statement, the learner does not have to use the provided numbered spaces. The vague reference to transport lipids is not incorrect, just insufficiently detailed to be creditworthy on its own, and so the list principle for marking does not apply here.*

(c) The RER and the smooth endoplasmic reticulum (SER) have different functions.

State **two** functions of the SER.

(2)

1 Synthesises, Stores and transports lipids.

Lead Examiner comment: *This response was awarded 0 marks. Package of carbohydrate is incorrect- we can only accept the idea of metabolism of carbohydrates. Transport is insufficient without clarification - such as transport of lipids from gut.*

(2)

1 package carbohydrates

2 transport in the cell.

Q3(a)

The majority of learners provided the correct answer.

A few multiplied the numbers, or got the sum upside down, despite being given the equation.

The most common errors arose from confusion over the unit's powers and squaring the surface area and cubing the volume and then carrying out the division.

Many learners missed out of the evaluation mark for 9892 / 14333 because despite correctly stating answer of this division as 3.3×10^{-4} they then shorted their final answer to 3.3.

Lead Examiner comment: *This response was awarded 2 marks. 0.69 is the correct response to 2 decimal places. We have not stated the number of decimal places and so any response which rounds up to 0.7 can be awarded maximum (2) marks.*

(2)

$$\text{SA to volume ratio} = \frac{\text{SA}}{\text{Volume}}$$

$$= \frac{989}{1433}$$

$$= 0.690$$

$$\approx 0.7 \text{ (1dp)}$$

surface area to volume ratio of the palisade cell = 0.7

Lead Examiner comment: *This response was awarded 1 mark. Due to the units given for surface area and volume unfortunately some learners have attempted to square, cube, square root, etc the values. ECF marks for a correct evaluation can be awarded, even though the substitution was incorrect.*

$$\frac{\sqrt{989}}{\sqrt[3]{1433}} = 2.78943395$$

surface area to volume ratio of the palisade cell = 2.79

Q3(b)

Almost all learners who provided an answer for this question achieved at least one mark with the majority scoring two or three. Most achieved marks for stating ‘absorb’ and naming a substance while some were also able to link an appropriate process to their substance or refer to increased surface area. There was little reference to channels or carriers suggesting learners may have limited knowledge about the plant cell structure and function, compared to animal cells.

Lead Examiner comment: *This response was awarded 3 marks. The learner has given a full explanation.*
On line 4 of their response, they state "absorb a larger quantity" which is sufficient for MP3. They then state of water, nutrients, and minerals- each would be sufficient for MP4, but each marking point can only be awarded once and so this gains the learner a further mark.
The learner then relates the water to photosynthesis- which is acceptable for MP6. Reference to light and photosynthesis can be ignored as this question referred to the root hair cell and so reference to light can be ignored.

The function of a root hair cell is to absorb water ~~and~~ nutrients and minerals from the soil. A large surface area to volume ratio means the root hair cell can absorb a large quantity of water and nutrients and minerals (a larger surface area provides a larger root hair to be embed and touch the soil). A larger quantity

(Total for Question 3 = 5 marks)

of water, nutrients and minerals allows the plant to photosynthesise more efficiently and have more energy. this larger amount comes from the larger surface area

Lead Examiner comment: *This response was awarded 0 marks. There is nothing creditworthy- the learner is unclear about the function of the root hair cell.*

The root hair cell gets rid of substances having a large surface area will unable them to ~~push~~ get rid substances more efficiently.
The root hair cell stops substances from getting inside the cell

Q4(a)

This question required specific knowledge of the sliding filament theory. due to the nature of a closed gap fill only specific responses were creditworthy. Only a small number of learners managed to achieve full marks on this question. Some learners referred to actin and myosin, suggesting knowledge of muscle contraction but they had not engaged with the details in the flow diagram. Some learners unfortunately confused ATP and ADP and did not notice that the missing word for B was connected to P_i.

Lead Examiner comment: *This response was awarded 3 marks. Cross bridge was also creditworthy for A.*

(3)

A cross-linking

B ADP

C ATP

This response was awarded 1 mark for C

A myofibrils

B [illegible] ATT

C ATP

Q4(b)(i)

The majority of learners achieved full marks here but, due to the format of the question, most who did not score 2 marks received no marks. The exceptions to this are those learners who gave the same answer for both boxes. Aside from getting the answers the wrong way round, the most common incorrect responses were references to speed ('fast' or 'slow') and 'yes'/'no'.

Lead Examiner comment: *This response was awarded 2 marks. Although the first word looks like "food" we can give benefit of the doubt from "or Low" and presume the first word is "poor".*
Therefore, each response is correct and both marks can be awarded.
Each box is worth a separate mark.

feature	fast twitch muscle fibre	slow twitch muscle fibre
level of blood supply	Poor or low	High
concentration of glycogen	High	Low
uses oxygen for respiration	No	Yes

Table 1

Lead Examiner comment: *This response was awarded 0 marks. The learner has given a final answer of high- low, which is incorrect and so no marks can be awarded.*

feature	fast twitch muscle fibre	slow twitch muscle fibre
level of blood supply	high ~~High low~~	~~low high~~ low
concentration of glycogen	High	Low
uses oxygen for respiration	No	Yes

Table 1

Q4(b)(ii)

Many learners found this question particular difficult to answer.
A few learners were able to identify the glycogen contains glucose.
However, a very common misconception was that fast twitch muscle fibres used glycogen instead of oxygen for respiration.

Not many learners stated that glycogen was broken down to glucose and few related the fact that because the glucose is only partly broken down, lots of glucose molecules need to be used in anaerobic respiration to give the quick energy release for fast twitch muscle fibres. Some referred to the fewer mitochondria, but some then went on to say that mitochondria are used for anaerobic respiration. Some linked the low blood supply to not needing glucose or oxygen delivered.

Many learners used the space to describe when fast and slow twitch fibres are used which didn't answer the question but tended to gain a mark for referencing fast twitch muscle fibres use anaerobic respiration. Often answers focussed on the type of activity which required anaerobic respiration and linked it to the high energy demand.
Some confused the terms anaerobic and aerobic – saying for example that no oxygen needed as they are aerobic or need oxygen for anaerobic. Some gave both fast twitch and slow twitch, which gave long answers but low marks. The reverse argument of any marking point can be accepted but the same marking point cannot be awarded twice.

Lead Examiner comment: *This response was awarded 4 marks. The learner has not given MP1- glycogen gives more energy is too vague and glucose being stored as glycogen is the wrong direction for the context of the question. The learner has used MP6 as their identification point- fast twitch has a high concentration of glucose is sufficient.*
In some responses we have seen that the learners are indicating that glycogen contains lots of glucose and is sufficient for both MP1 and MP6 and so would gain 2 marks from one short phrase. They have given sufficient for MP2 (anaerobic respiration)
MP3 can be awarded for the details about 2 ATP per glucose molecule.
The reference to low blood supply at the end of their response is related to fast twitch and so is correct and so a further mark for MP7 is given.

Fast twitch muscle fibres have a high concentration of glucose because they use aneorobic respiration which contain 2ATP molecules which isn't enough so the glucose stored -> glycogen gives more energy. ~~With the~~ Slow twitch fibres can go a long time without the muscles getting fatigued however fast twitch can't. They also have it because they dont use oxygen for respiration

(Total for Question 4 = 9 marks)

because of the low blood supply it has.

Lead Examiner comment: *This response was awarded 1 mark. The response does just give sufficient detail about glycogen containing glucose to be creditworthy- they have said glycogen can be turned into glucose which can be accepted for MP1.*

Fast twitch muscle fibres have a higher glycogen concentration because the muscles need lots of energy for the fast, repeating motions. So the glycogen is used so it can be turned into glucose for energy.

Q5

At Level 3, there were some excellent accounts with correct detail of the sodium potassium pump and ion movements by active transport, resting potential and depolarisation due to sodium ion channels opening, following a stimulus, and an influx/diffusion inwards of sodium ions down a concentration/electrochemical gradient, threshold with voltage gated sodium ions channels opening (some even referred to positive feedback) and the all or nothing responses that leads to an action potential.

Level 2 responses tended to realise the importance of the sodium-potassium pump in the resting potential, but often didn't get the detail surrounding the pump correct, such wrong ions going in/ out, numbers of ions, charge of those ions. Some thought sodium and potassium ions have a negative charge. Some thought that at depolarisation the voltage gated sodium ion channels shut (confusing it with reaching action potential).
Learners knew about the importance of sodium ions moving into the axon through channels but little reference to diffusion or moving down a gradient.

There were lots of answers making reference to the threshold potential, but few understood the significance of it in the 'All or nothing principle'.

Level 1 responses merely described the graph with very little detail given to explain the changes that happened in the graph and so the idea that the shape of the graph related to membrane permeability was relatively common.

Some learners referred to the potential difference values but some thought that going from -70 to -55 was a reduction in membrane potential rather than an increase, but some did correctly refer to the membrane becoming more positive or less negative.

Learners tended to say what is happening at points 1 and 2 rather than between points 1 and 2. The main misunderstanding here was going through the whole graph explanation rather than just from point 1 to point 2. This, of course, reduced the possibility of scoring well as they did not have the space to give fuller answers for the required section.

Lead Examiner comment: *This response was awarded 6 marks. This response shows comprehensive analysis of the scientific information in the figure. the response shows a well-developed structure, which is clear, coherent, and logical. They have clearly understood the question and have not given detail past point 2 (which would be ignored as we can only credit up to point 2 due to the scope of the question).*
There is no requirement to cover all of the indicative content to achieve maximum marks. This response includes correct reference to a number of the points in the indicative content. It shows a really good understanding of all the 3 stages of the indicative content- which certainly suggests a Level 3 response and we could give 6 marks for a less detailed response.

At point 1 the neuron is at resting potential, where no action potentials are being generated. However ions are still diffusing across the membrane at this point to maintain a negative potential inside the cell compared to outside the cell. Here the membrane is more permeable to potassium ions. For every 3 sodium ions that are pumped out of the neuron, 2 potassium ions are pumped in. This helps to maintain the negative membrane potential inside the cell compared to outside the cell.
At point 2 depolarisation is happening, where the membrane potential is becoming less negative. This is due to the fact that a nerve impulse is generated causing

the ~~sodium~~ voltage gated sodium channels to open. Therefore at this point the membrane is more permeable to sodium ions, and allows sodium ions to diffuse into the neuron down a concentration gradient. This increases the membrane potential above the -55 mV threshold, causing more sodium channels to open, further increasing the membrane potential as an action potential is generated.

Lead Examiner comment: *This response is a good example of a Level 1 response. The learner has mainly described the line on the graph and has not explained the changes in membrane permeability or membrane potential. This is creditworthy for 1 mark.*

(6)

at point 1 the ~~to~~ graph is steady at -70mv before an action potential in a neurone but during the act of potential in a neurone it increase ~~to~~ two 30mv at point 2 we are show it around -40 this is because during the an action potential in a neurone the membrane potential will increase as it is the body working.

Summary

Structure and functions of cells and tissues.

Based on their performance on this paper, learners should:

- Understand the demand of the command verbs, especially the difference between describe and explain.
- Be familiar with and recognise the ultrastructure and function of organelles in the following cells: prokaryote cells (bacterial cells), eukaryotic cells (plant and animal cells) and eukaryotic cells (plant-cell specific).
- Revise key definitions with Level 3 detail.
- Show their working out for all calculations to ensure marks can be given for the stages within the calculation even if the final answer is incorrect.
- Use the additional guidance to understand the depth and breadth of the specification.

The link to the Specification and/or Sample Assessment Materials (SAMs) is located on the BTEC Nationals qualification webpage here.

Individual Questions

Chemistry

Q1(a)(i)

This question asked learners to give one physical property of group1 metals.

Lead Examiner comment: *Many learners gave the answer that the metals also had low boiling points to gain the mark. Some knew the metals were soft to gain the mark as in this example.*

> 1 Lithium and sodium are metals in group 1 of the periodic table.
>
> (a) (i) A physical property of group 1 metals is that they have a low melting point.
>
> Give **one other** physical property of group 1 metals.
>
> (1)
>
> They are soft

Lead Examiner comment: *In other cases, learners gave a more generic physical property of metals, that they conduct heat or electricity to gain the mark.*

> 1 Lithium and sodium are metals in group 1 of the periodic table.
>
> (a) (i) A physical property of group 1 metals is that they have a low melting point.
>
> Give **one other** physical property of group 1 metals.
>
> (1)
>
> Good conductor of electricity

A noticeable proportion of learners did not read the question carefully and tried to give a chemical property and made reference to their reactivity or tried to explain why the group 1 metals are very reactive, neither of which gained credit.

Lead Examiner comment: *This example of a common incorrect answer scored 0 marks. Learners must be taught to read the question carefully, taking care to check the command word and ensure that the focus their answer appropriately.*

> 1 Lithium and sodium are metals in group 1 of the periodic table.
>
> (a) (i) A physical property of group 1 metals is that they have a low melting point.
>
> Give **one other** physical property of group 1 metals.
>
> (1)
>
> They have one electron in their outershell

Q1(a)(ii)

Learners were asked to complete a diagram of the metallic structure of a group 1 metal by adding the positive and negative charges.

Lead Examiner comment: *Some learners were able to complete the diagram correctly as in this example.*

(ii) Group 1 elements have a metallic structure.

Figure 1 shows an incomplete diagram of a metallic structure.

Complete Figure 1 by adding positive and negative charges to the diagram.

(2)

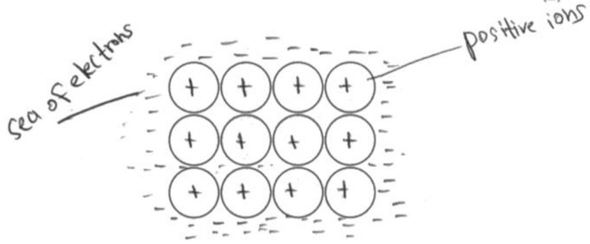


Figure 1

Lead Examiner comment: *However, a large number of learners confused the metallic structure with an ionic structure and so gained no marks.*

(ii) Group 1 elements have a metallic structure.

Figure 1 shows an incomplete diagram of a metallic structure.

Complete Figure 1 by adding positive and negative charges to the diagram.

(2)

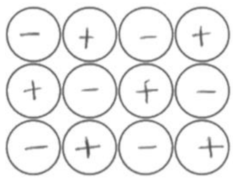


Figure 1

Q1(b)(ii)

Learners were asked to give a use of sodium hydroxide.

Lead Examiner comment: *Many knew that sodium hydroxide is used in 'cleaning' or cleaning products which gained the mark.*

(ii) Give **one** use of sodium hydroxide.

(1)

claning products.

In some cases, learners stated that a use of sodium hydroxide was glass or paper. Learners should be taught to be precise with their language and state that sodium hydroxide is used when making glass, paper or soap not used as glass, paper or soap.

Some learners just stated that sodium hydroxide was used in experiments or in the laboratory, this was considered to be too vague for credit.

Lead Examiner comment: *A large proportion of learners read the question too quickly and assumed that it said sodium chloride rather than sodium hydroxide and gave table salt as their answer, this gained no credit.*

(ii) Give **one** use of sodium hydroxide.

(1)

Table salt.

Q1(b)(iii)

This was the first calculation on the paper, asking learners to calculate the concentration of a solution of sodium hydroxide in g dm^{-3}. The question generally performed well with the majority of learners gaining both marks.

Example response

(iii) 1.50 g of sodium hydroxide was dissolved in water to make 0.25 dm^3 of sodium hydroxide solution.

Calculate the concentration of this solution of sodium hydroxide in g dm^{-3}.

(2)

Show your working.

$0.25 \div 1.50 = 0.16$

$1.50 \div 0.25 = 6$

concentration = 6 g dm^{-3}

Lead Examiner comment: *In some cases, learners did not read the question carefully and tried to calculate the concentration in mol dm^{-3}, rather than g dm^{-3}. Whilst this is a harder calculation of concentration, it does not answer the question posed and so therefore a mark of 1 was allowed.*

(iii) 1.50 g of sodium hydroxide was dissolved in water to make 0.25 dm^3 of sodium hydroxide solution.

Calculate the concentration of this solution of sodium hydroxide in g dm^{-3}. **(2)**

NaOH

Show your working.

Mr of sodium hydroxide = 23 + 16 + 1
= 40

$$\text{Mol} = \frac{1.50}{40} = 0.0375$$

$$\text{Concentration} = \frac{0.0375}{0.25} = 0.15$$

concentration = 0.15 g dm^{-3}

Lead Examiner comment: *In this example, the learner had their initial division upside down and so lost the first mark. In these cases, the second mark could have been awarded for the evaluation but unfortunately the learner has rounded incorrectly and so the evaluation mark could not be awarded. Learners should be taught to round their answer correctly when evaluating their answer.*

(iii) 1.50 g of sodium hydroxide was dissolved in water to make 0.25 dm^3 of sodium hydroxide solution.

Calculate the concentration of this solution of sodium hydroxide in g dm^{-3}. **(2)**

Show your working.

$$\frac{0.25}{1.5}$$

concentration = 0.16 g dm^{-3}

Q1(c)(ii)

This question proved more difficult for learners with few scoring all three marks available.

Lead Examiner comment: *In some cases, learners lost marks as they were not precise with their language. Referring to the loss of electrons plural rather than a single electron. The learner state that 'it' is closer to the nucleus but does not state what it is. Another common comment seen was that it is easier for lithium to lose an electron than gain which is true but does not answer the question. This answer therefore scored zero marks.*

(ii) Explain why the first ionisation energy of lithium is higher than the first ionisation energy of sodium.

(3)

the first ionisation is higher because of how reactive lithium is as it ~~[illegible]~~ closer to the nucleus and it's easier for them to loose an electron than to gain

(Total for Question 1 = 12 marks)

Lead Examiner comment: *Where learners scored a mark, it was often as they understood that the electron was harder to be lost or harder to removed in lithium than in sodium but could not explain that any further. Many simply stated that sodium had more electrons but did not refer to the extra shells in sodium or the resultant shielding effect of the extra shells. This example scored just 1 mark.*

(ii) Explain why the first ionisation energy of lithium is higher than the first ionisation energy of sodium.

(3)

- Lithium and sodium are in group 1.
- Lithium has a higher nuclear charge than sodium and is why it has a higher ionisation energy ~~because~~ therefore it is harder for an electron to be lost/ ~~gained~~ compared to sodium that has ~~less~~ ~~elect~~ ~~an~~ more electrons so it is easier for it to be lost.

(Total for Question 1 = 12 marks)

Lead Examiner comment: *Where learners did understand why the first ionisation energy of lithium is higher than sodium, they often gave a complete and detailed explanation, as in this example, to gain all three marks.*

(ii) Explain why the first ionisation energy of lithium is higher than the first ionisation energy of sodium.

(3)

Lithium has a higher first ionisation energy as it is harder to remove the valence electron. This is because lithium has a smaller atomic radius so the valence electron has more attraction to the nucleus. There is also less sheilding in lithium.

(Total for Question 1 = 12 marks)

Q2(a)(ii)

This question asked learners to balance the equation for the reaction between iron and bromine.

Lead Examiner comment: *This was well attempted with the majority of learners correctly balancing the equation to gain the mark.*

(ii) Add numbers to balance the equation for the reaction between iron and bromine.

(1)

..........2.. Fe + ...3.. $Br_2 \rightarrow 2FeBr_3$

Q2(b)

Learners found this question much more difficult with few scoring full marks. The question asked learners to explain how the electronegativity changes.

Lead Examiner comment: *Some learners tried to explain but did not include in their answer what the change was and so did not gain a mark for this. Explanations could still score, although many learners tried to explain in terms of the reactivity rather than the electronegativity of the elements. This example scored no marks.*

(b) Bromine and gallium are in period 4 of the periodic table.

The electronegativity of the elements in period 4 changes across the period.

Explain how the electronegativity changes from gallium to bromine.

(4)

Electronegativity is the tendancy to attract ~~the~~ electrons in a covalent bond. The electronegativity of elements change because the further along you move, the properties of the elements change. ~~The~~ From gallium to bromine, the electronegativity changes because bromine is more reactive and therefore ~~the~~ ~~not an al~~ doesn't have a electronegativity level.

(Total for Question 2 = 6 marks)

Q1(c)(ii)

Learners had a good understanding of electronegativity; they were able to give a full description to gain the full 4 marks available.

Example learner response

(b) Bromine and gallium are in period 4 of the periodic table. ·subshell·

The electronegativity of the elements in period 4 changes across the period.

Explain how the electronegativity changes from gallium to bromine.

(4)

Across ~~a~~ period 4, the electronegativity increases because as you go further, there is an extra proton within the nucleus attracting the electrons. The shielding is at a constant (primarily because they have the same outer sub-shell) However, as the electronegativity increases across period 4, the atomic radius decreases as there is a higher pull of electrons towards the nucleus. ~~As germanium and~~ ~~Germanium~~ Gallium has a lower amount of protons in the nucleus in comparison to Bromine, which is why Bromine is more electronegative than gallium.

(Total for Question 2 = 6 marks)

Q3(a)(ii)

This question asked learners to describe how a potassium atom forms a potassium ion.

Lead Examiner comment: *Answers to this question were variable with some learners being able to confidently and concisely explain how the ion is formed.*

(ii) Potassium ions, K^+, are formed from potassium atoms.

Describe how a potassium atom forms a potassium ion.

(2)

The potassium atom ~~an~~ needs to lose the last electron on its outer shell to become a potassium ion.

Lead Examiner comment: *In some cases, learners gave contradictions in their answers such as potassium loses and gains electrons or were not precise in their language and stated that potassium loses electrons plural rather than potassium loses one electron and so therefore missed out on a mark.*
This example gained just 1 mark for the understanding that the potassium atom loses electrons.

(ii) Potassium ions, K^+, are formed from potassium atoms.

Describe how a potassium atom forms a potassium ion.

(2)

When a potassium ~~ion~~ atom is oxidised it looses electrons & becomes more positive therefore creating a potassium ion. (K^+)

Lead Examiner comment: *In other cases, learners seemed less sure and used terms incorrectly therefore scoring no marks.*

(ii) Potassium ions, K^+, are formed from potassium atoms.

Describe how a potassium atom forms a potassium ion.

(2)

A potassium atom forms a potassiom ion by losing one of its atoms. ~~this means~~ Atoms are substances with one pure element. They have bonded to create an ion.

Q3(b)

The second calculation question on the paper asked learners to calculate the mass of potassium that would be needed from a known mass of potassium chloride.

Lead Examiner comment: *Where learners did well in this question, their answer was often set out in a logical way showing their understanding.*

(b) Potassium chloride can be made by reacting potassium with chlorine.

The equation for the reaction is

$$2K + Cl_2 \rightarrow 2KCl \quad 1.492g$$

Calculate the mass of potassium needed to make 1.492 g of potassium chloride.

(Relative formula mass of potassium chloride = 74.6)

(Relative atomic mass of potassium = 39.1)

moles = mass/mR.

(3)

Show your working.

1.492 74.6.

$\frac{1.492}{74.6} = 0.02$ there are 0.02 moles.

The reacting ratio is 2:2 so 1:1

$0.02 \times 39.1 = 0.782$

mass of potassium = 0.782 g

(Total for Question 3 = 6 marks)

Lead Examiner comment: *In some cases, learners used a different method to calculate the mass, which were accepted.*

(b) Potassium chloride can be made by reacting potassium with chlorine.

The equation for the reaction is

$$2K + Cl_2 \rightarrow 2KCl$$

Calculate the mass of potassium needed to make 1.492 g of potassium chloride.

(Relative formula mass of potassium chloride = 74.6)

(Relative atomic mass of potassium = 39.1)

(3)

Show your working.

2k + Cl₂ → 2kCl

74.6×2
= 149.2
×0.01
1.492

2k × 0.01 = Ar
k = 39.1 ×2 =
= 78.2
×0.01
0.782

mass of potassium = 0.782 g

(Total for Question 3 = 6 marks)

Lead Examiner comment: *In some cases, learners only completed part of the method and gained just 1 mark for calculating the number of moles of potassium chloride. In these cases, 1 mark was awarded as in this example.*

(b) Potassium chloride can be made by reacting potassium with chlorine.

The equation for the reaction is

0.02 moles

$$2K + Cl_2 \rightarrow 2KCl$$

Calculate the mass of potassium needed to make 1.492 g of potassium chloride.

(Relative formula mass of potassium chloride = 74.6) mr

moles = mass / mr

(Relative atomic mass of potassium = 39.1) ar

(3)

Show your working.

$$\text{moles KCl} = \frac{1.492}{74.6} = 0.02$$

mass of potassium = ... g

(Total for Question 3 = 6 marks)

Q4

This was the levels based 6 mark question. Learners were asked to explain, in terms of oxidation and reduction, observations seen during two reactions. Many learners knew that the iron nail oxidised in the copper(II) sulfate solution to gain a mark, as in this example. A large proportion did also think that the iron nail rusted. This was ignored.

4 An iron nail and a copper coin are placed in separate solutions of copper(II) sulfate.

The nail and the coin are removed from the copper(II) sulfate solutions after one hour.

Table 2 shows the recorded observations.

object	observations
iron nail	brown coating, solution loses blue colour
copper coin	no coating, solution remains blue

Table 2

Explain, in terms of oxidation and reduction, the observations shown in Table 2.

You may include equations to support your answer.

/reacted **(6)**

- The iron nail oxidised' in the copper (II) sulfate solution and started to rust (brown coating).

Lead Examiner comment: *This example scored 4 marks in Level 2. The learner has given the overall symbol equation, including state symbols for the reaction and have explained that, in the reaction, iron is oxidised from 0 to +2 and that copper has been reduced from +2 to 0.*

4 An iron nail and a copper coin are placed in separate solutions of copper(II) sulfate.

The nail and the coin are removed from the copper(II) sulfate solutions after one hour.

Table 2 shows the recorded observations.

object	observations
iron nail	brown coating, solution loses blue colour
copper coin	no coating, solution remains blue

Table 2

Explain, in terms of oxidation and reduction, the observations shown in Table 2.

You may include equations to support your answer.

(6)

$Fe_{(s)} + CuSO_{4(aq)} \rightarrow FeSO_{4(aq)} + Cu_{(s)}$

• Iron is ~~reduced~~ oxidised from 0 to +2

• Cu is reduced from +2 to 0

• Iron when reacting with copper(II) sulfate cause the Iron to rust, turning into brown colour causing solution to look less blue

•

Lead Examiner comment: *This last example scored 6 marks in Level 3. The learner has given a full explanation, in terms of loss and gain of electrons and oxidation states, the observations shown for both of the reactions.*

4 An iron nail and a copper coin are placed in separate solutions of copper(II) sulfate.

The nail and the coin are removed from the copper(II) sulfate solutions after one hour.

Table 2 shows the recorded observations. $Fe + CuSO_4 \rightarrow FeO +$

object	observations
iron nail	brown coating, solution loses blue colour
copper coin	no coating, solution remains blue

$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

Table 2

Explain, in terms of oxidation and reduction, the observations shown in Table 2.

You may include equations to support your answer.

(6)

- Oxidation is the loss of electrons and reduction is the gain of electrons
- Within the copper sulfate solution, there are Cu^{2+} and SO_4^{2-} ions
- The dissolved Cu^{2+} ions give the solution its blue colour
- When the iron nail was placed into the solution, the iron was ~~oxid~~ oxidised to form ions within the solution.
- A redox reaction occurred which is where oxidation and reduction occur simultaneously
- The electrons lost by the iron were gained by Cu^{2+} ions in the solution, known as reduction. This formed solid copper which built up a layer on the iron nail over time
- $Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$
- The ionic equation for the redox reaction is: $Fe + Cu^{2+} \rightarrow Fe^{2+} + Cu$
- $FeSO_4$ is colourless and therefore this explains why the solution loses its colour because the $CuSO_4$ becomes $FeSO_4$
- This occurs because copper is more reactive than iron - it is a displacement reaction
- When the copper coin is placed into the solution there is no reaction

because the metals have the same reactivity and therefore a displacement reaction doesn't occur.

$Cu + CuSO_4 \rightarrow X$

(Total for Question 4 = 6 marks)

TOTAL FOR SECTION B = 30 MARKS

Summary

To improve in future exam series learners should ensure that they understand what is required from different command words such as explain and describe. They should ensure they are specific with their answers and careful and precise with their use of scientific language.

They should be taught correct methodologies to calculations that come up in the unit and practice setting out their work in a logical manner so that they can arrive at a given answer in a concise way. They should be taught how to round their answer in their final evaluation correctly.

Individual Questions

Physics

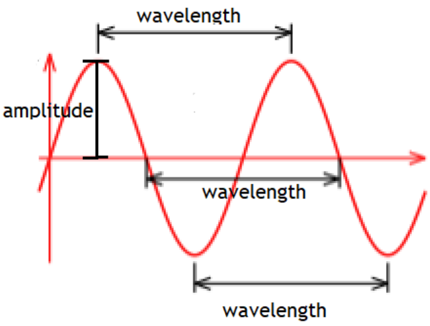
Q1(a)(i)

The amplitude being the distance between the between mean value and the maximum displacement was recognised by most learners.

Q(a)(ii)

A little more than half of the learners recognised the length that constituted a wavelength. The wavelength is the distance between two consecutive troughs or crests, this is the easiest to remember. If the central zero amplitude line is used it is the distance between two successive points on the wave which are moving in the same direction.

Lead Examiner comment: *The measurements to be taken for amplitude and wavelength are shown below:*



Q1(b)

Almost all learners were able to substitute into the given equation and were able to evaluate the speed of the wave.

Example response

(b) A different transverse wave has a wavelength of 1.8 m and a frequency of 16.0 Hz.

Calculate the speed of this wave.

Use the equation: $v = f\lambda$

(2)

$v = f\lambda$ $\quad f = 16$

$v = 16.0 \times 1.8$ $\quad \lambda = 1.8$

$v = 28.8$

speed of wave = 28.8 ms^{-1}

Lead Examiner comment: *A few learners did not recognise that v=fλ means v = f x λ and did not complete the substitution correctly.*

Lead Examiner comment: *The example below gained no marks as the substitution was incorrect and the value obtained was not rounded correctly. The calculation gives 8.888(recurring) which should have been rounded to 8.9.*

(b) A different transverse wave has a wavelength of 1.8 m and a frequency of 16.0 Hz.

Calculate the speed of this wave.

Use the equation: $v = f\lambda$

(2)

$v = \frac{16}{1.8} = 8.8$

speed of wave = 8.8 ms^{-1}

Q1(c)

The relationship between wavelength and phase difference was recognised by about half of the learners who understood that one wave is equivalent to 360° and therefore a phase difference of ¼wavelength is 90°.

Q1(d)

Again, about half of the learners were able to gain both marks. Many more knew that the answers were ‘perpendicular’ and ‘parallel’ but were confused about which was which. The use of the correct terms should be encouraged from the start of teaching this topic as the use of descriptions such as ‘side to side’ without any further explanation can be applied equally to transverse and longitudinal waves.

Q2(a)(i)

Almost all learners were able to use the graph to convert $\frac{5}{8}$V to the digital code 101.

Q2a(ii)

Many learners were able to gain one mark for either giving an identification or an expansion. However, to gain both marks the identification and expansion need to be linked. The most common one mark answer was that' digital signals travel further' to gain two marks the learner needs to give the reason for this which is that 'the signals can be regenerated'.
Some examples of two mark answers are given below.

Example response 1

(ii) Information can be transmitted as a digital signal or an analogue signal.

Explain **one** advantage of transmitting information as a digital signal.

(2)

There is no data loss because a digital wave is either at 1 or 0 unlike an analogue signal

Lead Examiner comment: *The identification is 'prevents data loss' expansion 'uses binary code'.*

Example response 2

(ii) Information can be transmitted as a digital signal or an analogue signal.

Explain **one** advantage of transmitting information as a digital signal.

(2)

as a digital signal

one advantage of transmitting information is that it can be encrypted by security therefore when used for communication the information is safe and secure can not get leaked.

Lead Examiner comment: *The identification is that the 'signal is encrypted' and the expansion is that ' information is safe and secure and not leaked'.*

Q2(b)

Only a few learners understood that to access the internet there needs to be intermediary between the computer and the internet and this is commonly a ‘router’ although a wide variety of intermediate devices were credited. The way these connections are made was credited for the third marking point.

The example below gained all three mark

(b) A person switches on a computer and the computer accesses the internet.

Describe the process a computer uses to access the internet through a Wi-Fi connection.

(3)

the computer ~~interacts~~ connects ~~with~~ to the wi-Fi router, then the Wi-Fi router interacts with a sattelite within the ionosphere, the sattelite connects the router to the internet and the computer accesses the internet through the router.

Q3(a)

This was the most challenging of the multiple chose questions with less than half of the learners able to identify from the diagrams that a ray of light striking a glass –air boundary at the critical angle will be refracted through 90°.

Q3(b)

Learners did best if they had a ruler to complete the diagram this is a requirement for the examination. A measurement of the angle between the reflected ray and the top surface of the optical fibre was not needed as it was judged by eye. There were a number of learners that drew a ray refracted out of the optical fibre and received no credit. About half of the learners could gain one mark by drawing a reflected ray regardless of the angle and those learners that made the angles look about equal gained the second mark.

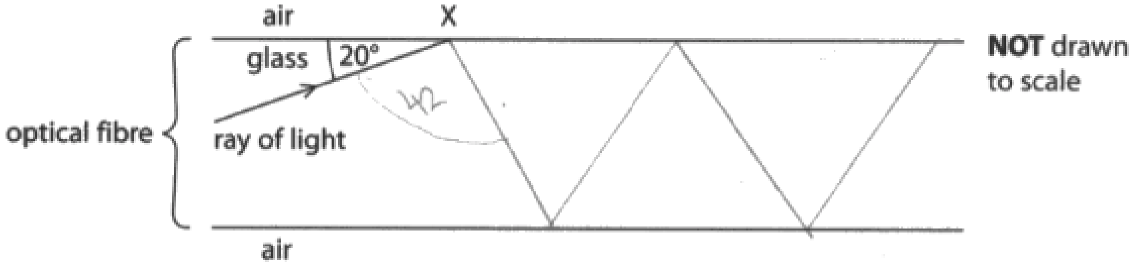
Lead Examiner comment: *The example below gains 1 mark for correctly recognising that there is a reflection but ignoring the angle.*

(b) Figure 6 shows a ray of light inside part of an optical fibre.

The critical angle for this glass-air boundary is 42°.

Draw a line on Figure 6 to show the path of the ray of light from point X as the ray of light travels through the optical fibre.

(2)



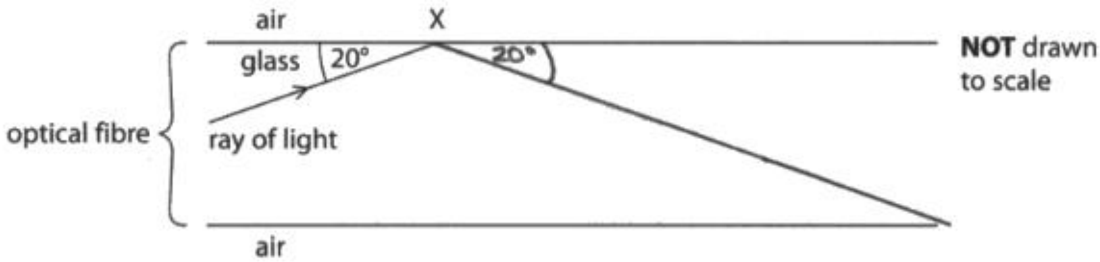
Lead Examiner comment: *The example below gains two marks. Reflection is shown and at about the right angle.*

(b) Figure 6 shows a ray of light inside part of an optical fibre.

The critical angle for this glass-air boundary is 42°.

Draw a line on Figure 6 to show the path of the ray of light from point X as the ray of light travels through the optical fibre.

(2)



Q3(c)

Surprisingly only just over half of the learners were able to identify a medical application of optical fibres. The most common answer was ‘endoscope’ closely followed by keyhole surgery. Those that did not know the medical terms could gain a mark from ‘see inside the body’.

Q4(a)

This question was the most challenging on the paper as the explanation required the link between frequency and wavelength to be used and the fact that the velocity of sound would be the same in both pipes. The diagrams show the difference in wavelength of sound in a closed and open pipe and just noting this difference would have gained one mark. Learners need to use the information in the diagrams and not just read the text as diagrams give much useful information which can be used to answer the question.

The example below gives a rare correct 3 mark answer

4 (a) A learner has two pipes of the same length, L.

One pipe is open at both ends and the other pipe is closed at one end.

Air is blown into each pipe to produce a standing sound wave.

The standing wave gives a sound of specific frequency (pitch).

Figure 7a shows the standing sound wave in the open pipe.

Figure 7b shows the standing sound wave in the closed pipe.

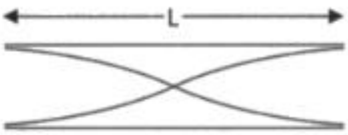


Figure 7a

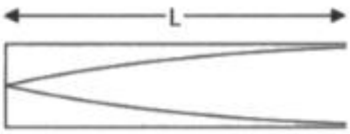


Figure 7b

Explain why the closed pipe in Figure 7b produces a sound with a lower frequency.

(3)

The length of the pipe in Figure 7b is 3/4 a wavelength of the standing wave, Figure 7a's wave has a wavelength of 2L. 7a's wavelength is shorter than 7b's. As such, the frequency is higher because the speed of the wave only depends on its medium and λf = V. Longer wavelengths mean lower frequency.

Q4(b)

The calculation required the use of standard form, rearranging, squaring and then evaluation by taking a square root. The majority of learners attempted the question and did at least put in values for the substitution although many did not recognize the relevance of the subscripts that is I_1 was the intensity for the distance D_1 often one side of the equation was inverted. Those learners that were able to correctly substitute and rearrange were usually able to complete the calculation and gain all four marks.

Lead Examiner comment: *The example below shows that the learner has identified the values to be used, substituted in the equation, rearranged and calculated a value for $(D_2)^2$.*
A square root has then been taken to find D_2.

(b) The intensity of a sound produced by a loudspeaker varies with the distance from the loudspeaker.

The intensity of the sound at a distance of 2.5 m from the loudspeaker is 1.0×10^{-5} Wm^{-2}.

Calculate the distance from the loudspeaker where the intensity of sound is 5.0×10^{-10} Wm^{-2}.

Use the equation:

$$\frac{I_1}{I_2} = \frac{(D_2)^2}{(D_1)^2}$$

(4)

$$\frac{1.0 \times 10^{-5}}{5.0 \times 10^{-10}} = \frac{(D_2)^2}{(2.5)^2}$$

$$\frac{1.0 \times 10^{-5}}{5.0 \times 10^{-10}} \times (2.5^2) = (D_2)^2$$

$$(D_2)^2 = 125000$$

$$\sqrt{125000} = 353.5533906$$

354 to 3 sf

distance from loudspeaker = 354 m

Q5

The question was about microwaves used in mobile phone communication and not about microwaves used for other purposes. The most common misunderstanding was that mobile phones can be used over long distances. Mobile phones have to be within range of a base station for their network in order to send or receive a signal. Mobile phones do not communicate with satellites. Any mention of satellite communication, health and safety or microwave ovens was regarded as irrelevant.

Lead Examiner comment: *About half of the learners were able to gain a Level1 by giving some advantage or disadvantage of using microwaves for mobile phone communication but without detail. Generic statements were made, and the lines of reasoning were generally unsupported.*

Example response Level 1

5 Mobile phones use waves with a range of frequencies in the microwave region of the electromagnetic spectrum.

Compare the advantages and disadvantages of using microwaves for mobile phone communication.

(6)

The advantage of using microwaves for mobile phone communication is that they are faster for connection and easy to use.
-the disadvontage is that the connection com be slower due to weather or being in a very closed area like high up buildings were you can't get connections.

Lead Examiner comment: *About a quarter of learners were able to achieve Level 2. In the example below although the learner has not exactly answered the question in referring to radio waves and infrared there are sufficient advantages that have been supported by relevant evidence to award a Level 2.*

Example response Level 2

5 Mobile phones use waves with a range of frequencies in the microwave region of the electromagnetic spectrum.

Compare the advantages and disadvantages of using microwaves for mobile phone communication.

(6)

Microwaves have a higher frequency than radio waves, allowing more information at once.

Microwaves have shorter wave lengths though.

microwaves have longer wave lengths than infrared waves.

They dont need a direct line of sight unlike infrared and visible light so they can travel through objects.

Radio waves are lot more likely to have interference.

Microwaves can ~~[illegible]~~ give more information within a period of time compared to ~~[illegible]~~ radio waves.

Lead Examiner comment: *A few learners were able to achieve Level 3 by presenting both advantages and disadvantages and giving lines of argument that were consistently supported and maintained by the application of relevant evidence.*

Example response Level 3

5 Mobile phones use waves with a range of frequencies in the microwave region of the electromagnetic spectrum.

Compare the advantages and disadvantages of using microwaves for mobile phone communication.

(6)

The disadvantage of using a microwave are that the signals can be lost this could be because of mountain or hills, large building which are interfering with it and this would mean that information would get lost between the signals. This can also occur because of bad weather such as rain or snow or even because of floods. Another disadvantage of using microwaves for mobile phone communication is that it is not secure which means information could easily be accessed and hacked aswell as information getting lost.

An advantage of this is that it can carry a lot of information and has a high bandwidth + frequency which means information can be clear.

Summary

To improve their mark for this paper learners should:

- Always show substitution into an equation and show working.
- Learn the meaning of symbols and how the symbols are presented in equations.
- Learn to rearrange equations.
- Take a calculator and ruler into the examination.
- Read the labelling on diagrams and use the information given in diagrams.
- Learn to mark a wavelength on a wave diagram.
- Learn the difference in particle movement for transverse and longitudinal waves.
- Read the questions carefully and actually answer the question.